

Watcher v2 — Build Roadmap

From a message filer to a receptionist. Every item scored for urgency and ease. • 15 August 2026 • v2.1

The correction this revision makes: v1.15 said 5.75 days remained. That counted only the numbered Track 0–3 items — it did not count the code that makes any of them execute. There is **no LLM client, no database connection, no application endpoint, no container and no frontend**. The receptionist can be reasoned about; it cannot yet process a single real message. A second pass then read intents.yaml against the schemas and the design spec, and found guardrails the vocabulary requires but nothing enforces. Honest remaining: **~43.75 engineering days**.

Where we stand today

Built and tested — the business logic is genuinely good	Missing — and nothing runs without these
277 passing tests, no DB or network needed 74 source files across 15 modules (106 with tests) 19 DB tables + 3 migrations Every external system behind a swappable seam Eval runner + CI accuracy gate, 50 golden cases Receptionist vocabulary: 19 intents, data not code Unified intent taxonomy — classifier = vocabulary Prompt v3 — catalogue rendered from the vocabulary, tie-breaks, language rules, injection rule Channel-neutral envelope + chat/voice adapters Task state machine and the autonomy gate Persistence layer — conversations, turns, tasks Channel-neutral core — KNOWN_LEAKS = {}, enforced by test	No LLM client — provider.py is a Protocol; zero HTTP calls exist anywhere in apps/ No DB connection — no engine or sessionmaker outside tests No endpoint — create_app() has no production caller No outbound — ChannelSender has no implementation No Dockerfile — so cd.yml has never deployed anything No RLS — no policy in any migration No control page — one CSS file, no package.json No control-page API — ~25 endpoints, none written Knowledge — zero tables, zero rows Live availability — no read path to any PMS A measured prompt — the gate replays v2 fixtures, so 88% is not v3's number

What changed in this revision

v1.15 → v2.1 (audit + design review)	What did not change
Five unpriced tracks added: A make it run, B host it, D control page, E product surface, G guardrails Total restated: 5.75 → 35.25 → ~43.75 days 2.1 / 2.2 downgraded — conversation continuity is built but never wired; every message is treated as turn one Emergency detection is never invoked — a gas leak files a maintenance ticket today Control page rebuilt — six receptionist views; the old five serve the v1 message-filer New item 2.8 — many properties per client, not one Decisions D14–D26 locked in DECISIONS.md	The business logic is sound and stays as built 2.4 knowledge is still the last item before a demo 3.1 still blocked on P1 (pick the first client) The scope guard still holds — no PDF handbook ingestion

Totals: ~43.75 engineering days remaining. At the observed rate of 2–3 engineering days per working session, that is **15–20 sessions**; at the six-day week these estimates assume, ~7.3 weeks of pure build, so plan **9–10 weeks to sellable** and **~13.5 days to a real number that answers safely**. The critical path is no longer 2.4 — it is **Track A**, because until the app can start and call a model, no other item can be demonstrated at all.

Track A — Make it run. [THE NEW CRITICAL PATH]

None of this is on the old roadmap, and nothing else can be shown until it exists.

#	Work item	Urgency	Ease	Days	Status
A1	Configuration layer	NOW	Easy	0.5	NOT BUILT — pydantic-settings over all ~20 vars in .env.example; only 2 are read today
A2	DB engine and session	NOW	Easy	0.5	NOT BUILT — engine, sessionmaker, get_session. Supabase transaction pooling needs NullPool + prepare_threshold=None
A3	Concrete LLM providers	NOW	Moderate	1.5	NOT BUILT — Anthropic + OpenAI against the existing LLMProvider seam. Mark the ~5k-token system block cacheable or it is the largest cost line
A4	Composition root	NOW	Moderate	1.0	NOT BUILT — apps/api/main.py, the first production caller of create_app()
A5	Wire continuity + excise v1 filer	NOW	Moderate	2.25	PART-BUILT — ConversationRepository is never called outside tests; the orchestrator passes task=None and empty slots. Also converts process() to async, removing the asyncio.run() crash, and removes the rules/destinations threading (D24) while the surgery is already open. Tables retained, not dropped
A6	WhatsApp outbound sender	NOW	Moderate	1.0	NOT BUILT — ChannelSender has no implementation; replies are composed and never sent

Why A5 is not a small item. Items 2.1 and 2.2 were marked DONE and both are genuinely written — the tables, the repository, the task state machine, the reply path. What was never written is the line joining them. worker.py calls the receptionist with extracted_slots={} and task=None hardcoded, so slot filling and task continuity are inert. The milestone “holds a conversation” is not yet true.

Track B — Host it. [Supabase + Render]

Stack locked 15 August 2026. B2 must land before a second client exists.

#	Work item	Urgency	Ease	Days	Status
B1	Supabase project and migrations	NOW	Easy	0.5	NOT STARTED — provision, set DATABASE_URL, run the 3 existing migrations
B2	Row-Level Security	NOW	Moderate	1.5	NOT BUILT — no policy in any migration, though AGENTS.md calls tenant isolation non-negotiable. Per-session GUC + a cross-tenant read test
B3	Dockerfile and Render service	HIGH	Easy	0.75	NOT BUILT — the Dockerfile alone activates the dormant cd.yml image job
B4	Domain, TLS, Meta webhook	HIGH	Easy	0.5	NOT STARTED — Meta needs a public HTTPS URL before anything arrives
B5	Durable queue (Redis + arq)	MED	Moderate	1.0	NOT BUILT — BackgroundTasks loses in-flight messages on every deploy. The seam already exists, so nothing else changes

Track 2 — Make it a receptionist. [2.4, 2.7, 2.8 REMAIN]

The reply path is written and the prompt now contains instructions a model can follow. What remains is knowledge, a measured prompt, and more than one property.

#	Work item	Urgency	Ease	Days	Status
2.1	Conversations, tasks and slot filling	HIGH	Hard	spent	PART-BUILT — 2.0d already spent: 7 tables, ConversationRepository, task converters, all written and tested. Not wired into the pipeline. The wiring is priced in A5, not here — do not add this row to the total
2.2	The reply path	HIGH	Moderate	spent	PART-BUILT — 1.5d already spent: tool registry, receptionist, ChannelSender protocol. No sender implementation, so nothing is delivered. Priced in A6, not here
2.3	Autonomy gate	HIGH	Easy	1.0	DONE — RECEPTIONIST_REPLY as the fourth RoutingAction; decide_autonomy() runs after classification and identity, before rule matching
2.4	Knowledge base	HIGH	Moderate	2.0	NEXT — facts table with sensitivity flags, a real “I don’t know” that fetches a human, verification codes
2.5	Prompt v2 and rewrite the golden set	MED	Moderate	1.0	DONE — golden set 8 → 50 across all 19 intents. The prompt half was replaced by v3
2.6	Intent taxonomy unification	NOW	Easy	0.5	DONE — 19 vocabulary-aligned intents
2.7	Re-record the eval under prompt v3	MED	Easy	0.5	PENDING — unblocked the moment A3 lands and a key exists. Add Franco-Arabic cases
2.8	Many properties per client	HIGH	Moderate	1.0	NEW — the roadmap assumed one property whose facts fit in the prompt. Needs a properties table, per-property fact scoping, and message→property resolution

The eval is still not measuring the prompt. AGENTS.md says no prompt change merges without an eval run. The gate does run and does pass — but it replays fixtures keyed by message text and recorded under prompt v2, so it reports 88% whatever the prompt says. Until 2.7 runs, **treat 0.88 as v2’s number.** baseline.json now says so in the file.

Track D — The control page. [SIX RECEPTIONIST VIEWS]

DESIGN-SPEC §8 specifies Inbox / Sources / Destinations / Rules / Admin — the UI of the v1 message-filer, where the job is triaging a record and routing it to a Sheet. The v2 receptionist has nine tools and talks to guests. Two of those five views serve a product we are no longer building.

#	Work item	Urgency	Ease	Days	Status
D0	Rewrite DESIGN-SPEC §8	NOW	Easy	1.0	NEW — new view set against the nine tools. Tokens, type scale, confidence chip and the RTL rules in §9/§10 all carry over unchanged
D1	Next.js scaffold, tokens, fonts, Clerk	HIGH	Easy	1.0	NOT STARTED — apps/control-page holds one CSS file. Committing package-lock.json activates CI's dormant web job
D2	REST API behind the views	NOW	Hard	3.0	NOT BUILT — the hidden half of this track. Tenant-scoped, paginated endpoints for handoffs, conversations, emergencies, properties, quotes and eval. None exist today
D3	Typed client from the OpenAPI schema	MED	Trivial	0.25	NOT STARTED — generated, not hand-written
D4	Handoff queue	NOW	Moderate	2.0	NEW — the operator's work queue and the replacement for Inbox: everything the bot escalated, with why . One reason per item, which is why D24 removes the second escalation mechanism
D5	Conversations	HIGH	Moderate	1.5	NEW — live threads, what the bot said and on what basis, take-over by a human
D6	Emergencies	NOW	Easy	0.75	NEW — alarm-like, acknowledged, never buried in a queue. Backed by G3
D7	Properties & Facts	HIGH	Moderate	1.5	NEW — the knowledge editor that makes Track C usable, sensitivity flags visible on the row
D8	Quotes & Audit	HIGH	Moderate	1.0	NEW — every price the bot said, with provenance, re-checkable months later. Also where the D20 door-code risk is recorded and visible
D9	Admin / Eval viewer	MED	Easy	0.75	NOT STARTED — accuracy, per-language breakdown, spend. Founder-only
D10	Arabic / RTL and accessibility	HIGH	Moderate	1.0	NOT STARTED — DESIGN-SPEC §9 and §10 carry over, applied across the new views

Track G — Receptionist guardrails. [THE VOCABULARY ALREADY REQUIRES THESE]

Not new features. These enforce rules intents.yaml states and schema.py validates, and which nothing in apps/ currently honours.

#	Work item	Urgency	Ease	Days	Status
G1	Quote path and provenance	NOW	Moderate	2.0	NOT BUILT — AvailabilityResult has no rate_or_quote_id and no valid_until, so as shipped it cannot satisfy quoting.provenance_required , which schema.py:176 enforces. Two modules contradict each other. Adds the 300s freshness bound and the audit write that must land before the number reaches the guest
G2	Reservation lookup + verification	HIGH	Moderate	1.5	NOT BUILT — booking reference plus a second fact (D19). Fixes the identity_verified wiring in the same pass so a fuzzy name match stops reading as proof

#	Work item	Urgency	Ease	Days	Status
G3	Emergency path	NOW	Moderate	1.5	NOT BUILT — worker.py:162 hardcodes emergency=False, so a gas leak files a maintenance ticket. Wire trigger matching before classification, then Twilio call to the operator
G4	Client config split	MED	Easy	0.5	NEW — safety-critical keys stay in YAML behind the build validator; currency, timezone, hours and wording move to the DB (D23)

The door-code decision (D20), with its reasoning. Door codes unlock on booking reference plus a second fact. The case for it: about half of bookings are OTA, the platform has already verified the guest by account and payment, and that guest gets the code through the OTA app anyway. Residual risk accepted: both facts appear on the confirmation email. Note the trap — requiring the sender's number to match the reservation looks like the fix but **fails for OTA guests precisely**, because Airbnb and Booking.com issue masked relay numbers (SESSION-HANDOFF §8: about half of all guests), and Reservation carries no phone field. **The one to revisit (D20a) is lock type:** a per-booking PIN expires at checkout, a leaked static key-box code compromises every future guest until someone drives out to the property.

Track E — What makes it sellable rather than working

Needed because the goal is a product with several paying clients, not a pilot.

#	Work item	Urgency	Ease	Days	Status
E1	Tenant onboarding	HIGH	Moderate	1.0	NOT BUILT — create a tenant, connect a number, seed properties, without touching SQL
E2	Usage metering and limits	MED	Easy	0.75	NOT BUILT — the usage_events table exists; nothing writes to it
E3	Billing (Stripe)	MED	Moderate	1.5	NOT STARTED — droppable to zero if the first clients are invoiced manually
E4	Observability	HIGH	Easy	0.75	NOT STARTED — structured logs, Sentry, uptime checks
E5	Backups and residency statement	MED	Easy	0.5	NOT STARTED — a restore drill, and the residency story the regulated tier is sold on

Track 3 — Integration and measurement

Base360.ai is ruled out as a partner. Hostaway, Guesty and Cloudbeds all publish APIs.

#	Work item	Urgency	Ease	Days	Status
3.1	PropertySystemPort plus the first adapter	MED	Moderate	2.5	PENDING — the published /v1/property-system router is an inbound projection; it fetches nothing. D16: an outbound HttpPropertySystemAdapter (~1.0) plus the Hostaway bridge, which Watcher writes and hosts (~1.5). Vendors will not implement your contract, so the neutral port relocates the adapter rather than removing it
3.2	End to end on a real number, then measure	HIGH	Moderate	1.0	PENDING — the point at which the eval number becomes real rather than recorded
3.3	Tuning tail	MED	Hard	2.0	PENDING — getting to working is weeks away; getting to trustworthy is measured, and that number is not fully under your control

Runs in parallel — start today

None of these are engineering. All of them can quietly become the reason a date slips.

#	Work item	Urgency	Ease	Days	Status
P1	Pick the first client	NOW	Easy	--	NOT STARTED — decides which PMS adapter gets built first. Without it, 3.1 is a guess
P2	PMS sandbox keys	HIGH	Easy	0.5	NOT STARTED — public docs, but approval and rate limits are theirs to grant. Shape the port from what two or three of them offer in common
P3	File Meta business verification	HIGH	Easy	1 hr	NOT STARTED — v1.15 called this no longer blocking. At paid volume across several clients it binds again. File it now
P4	Graphify as a build aid (optional)	LOW	Easy	0.5	NOT STARTED — maps the repo for the coding agents. Local parsing, nothing leaves the machine

How the total reconciles. A 6.75 + B 4.25 + D 13.75 + E 4.5 + G 5.5 + (2.4, 2.7, 2.8) 3.5 + (3.1, 3.2, 3.3) 5.5 = **43.75**. Rows 2.1 and 2.2 show “spent” rather than a number because their days are already delivered and the outstanding wiring is priced in A5 and A6 — adding them would double-count 3.5 days. The parallel track P1–P4 is excluded: it is founder time, not engineering. **Counting scope loosely is what produced the 5.75 figure this document replaces**, so the arithmetic is stated rather than implied.

Completed tracks (kept for the record)

Track 0 — Today. [COMPLETE]

#	Work item	Urgency	Ease	Days	Status
0.1	Set default branch to main	NOW	Trivial	1 min	DONE
0.2	Remove client name from golden set and fixtures	NOW	Easy	0.25	DONE
0.3	Decide the receptionist intent vocabulary	NOW	Easy	1 hr	DONE — 19 intents, 84 examples, 5 languages, 38 tests
0.4	Merge the eval branch	NOW	Trivial	0.25	DONE
0.5	Delete the stale nifty-johnson branch	LOW	Trivial	1 min	DONE

Track 1 — Foundations. [COMPLETE]

#	Work item	Urgency	Ease	Days	Status
1.1	Stop the core speaking one channel	NOW	Moderate	1.5	DONE — external_id / thread_id / sender_display_name, channel field, migration 002, KNOWN_LEAKS = {} with a boundary test holding it there
1.2	Port the four kept scaffold files	HIGH	Moderate	1.0	DONE
1.3	Python 3.12 to 3.13	LOW	Easy	0.5	DONE — 7 version pins across pyproject, packages and CI

The dates (revised 15 August 2026)

Milestone	v1.15 said	Revised	Status
Eval merged, name leak fixed, vocabulary decided	Day 1	--	DONE
Core stops speaking one channel; scaffold ported	Week 1	--	DONE
Taxonomy unified, prompt v2, eval at 50	14 Aug	--	DONE
Prompt v3 — instructions a model can follow	14 Aug	--	DONE
M1 — answers a real message, safely	~2 days from 14 Aug	~13.5 days	NEXT — Track A + B1-B4 + G3 + 2.4. The emergency path is not optional here
M2 — you can watch and correct it	not scheduled	~+13.75 days	PENDING — Track D, six receptionist views
M3 — sellable	~5 days from 14 Aug	~+16.5 days	PENDING — Tracks E, F and the rest of G. Blocked on P1
Phone answering	Week 4	after M3	PENDING — the speech-to-text seam already exists

What would actually move these dates

The gap was never technical difficulty — it was scope accounting. Every item in Tracks A, B, D and E is ordinary, well-understood work. None of it is hard in the way 2.1 was hard. It was simply never on the list, so the list said 5.75 days while the system could not start.

Supabase connection pooling. Transaction-mode pgbouncer and SQLAlchemy prepared statements do not get along. Settle it in A2 with a load test, not in production at 2am.

RLS retrofitted late. Adding row-level security after application queries exist surfaces as silent empty result sets rather than errors. B2 sits before the second tenant for that reason.

D2 is invisible in its own track title. “Build the control page” reads as frontend work; three of its days are backend endpoints that do not exist. It is the item most likely to be cut by accident.

Cost per message is still unmeasured. Nothing has called a model, so there is no unit economics figure. A3 makes it measurable and prompt caching is what keeps it small. Measure before quoting a price.

PMS API access. Docs are public, but sandbox approval and rate limits are theirs to grant. Start P2 today; it is the new version of the Meta-verification lesson.

If you do only one thing next session: start Track A. Not 2.4. The knowledge base was the right answer while the question was “what does it lack?” — but the receptionist cannot start, cannot reach a model and cannot send a reply, and no amount of knowledge changes that. A1 through A4 is the shortest path to a system that exists — and G3 before any real guest can message it.

Session log

Session 1: PR #13. Tests 228 → 248 (+20). Items 1.1, 1.3, 2.1, 2.2, 2.3 = 6.5 engineering days.

Session 2: PR #14 and #15. Tests 248 → 259 (+11). Items 2.5, 2.6 = 1.5 days.

Session 3: PR #16. Tests 259 → 277 (+18). Prompt v3, ~0.5 days, unplanned.

Session 4: Code audit plus a design review reading `intents.yaml`, `property_system/schemas.py` and DESIGN-SPEC against each other. No application code changed. Five unpriced tracks added; 2.1/2.2 downgraded to part-built; emergency detection found unwired; the quote schema found unable to satisfy its own provenance rule; the control page rebuilt around six receptionist views. Decisions D14-D26. See `docs/LAUNCH-PLAN.md`.

Cumulative: 8.5 engineering days delivered. ~43.75 remaining.